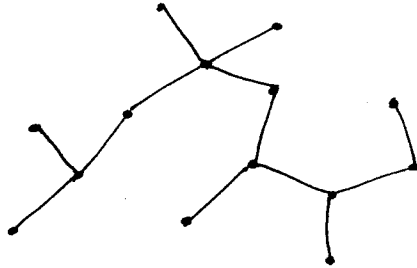


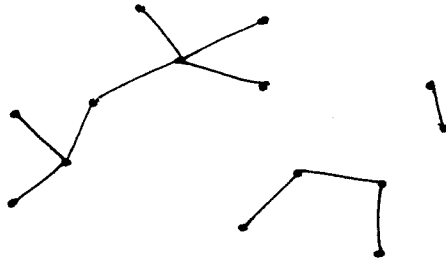
Appendix B.5: Trees

1) Free Trees: A free tree is a connected, acyclic, undirected graph. An undirected, acyclic graph that is disconnected is a forest. A graph that contains a cycle is neither a tree nor a forest.

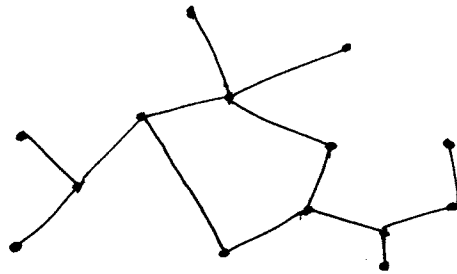
eg: a free tree



eg: a forest



eg: a graph that contains a cycle and hence is neither a free tree nor a forest



2) Properties of free trees:-

Let $G = (V, E)$ be an undirected graph.

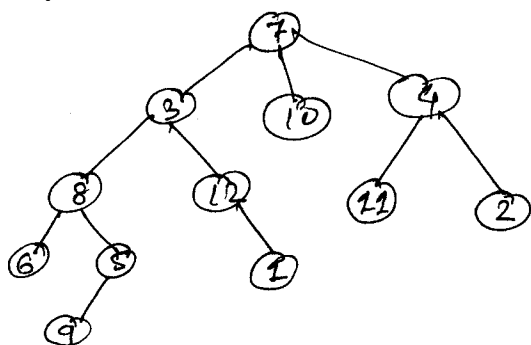
Then the following statements are equivalent:-

- 1) G is a free tree.
- 2) Any 2 vertices in G are connected by a unique simple path.
- 3) G is connected, but if any edge is removed from E , the resulting graph is disconnected.
- 4) $|E| = |V| - 1$
- 5) G is connected. G is acyclic.
- 6) G is acyclic, but if any edge is added to E , the resulting graph contains a cycle.

3) Rooted trees:-

A rooted tree is a free tree in which one of the vertices is distinguished from the others. The distinguished vertex is called the root of the tree.

eg: rooted tree



depth 0

depth 1

depth 2

depth 3

depth 4

↑
height = 4
↓

There is a unique path from the root r of a rooted tree to any node x . Any node on this unique path is called an ancestor of x (this includes x as well).

If y is an ancestor of x then x is a descendant of y . Every node is both an ancestor and a descendant of itself.

If y is an ancestor of x but $y \neq x$ then y is called a proper ancestor of x and x is called a proper descendant of y .

The immediate ancestor of a node is called its parent. If y is a parent of x then x is called a child of y . If two nodes have the same parent, they are siblings. A node with no child is called an external node or a leaf. A non-leaf node is called an internal node.

Degree of a node in a free tree: is the number of adjacent vertices.

Degree of a node in a rooted tree: is the number of children of the node (the parent of a node does not count toward its degree).

The length of the path from the root to a node is called the depth of the node.

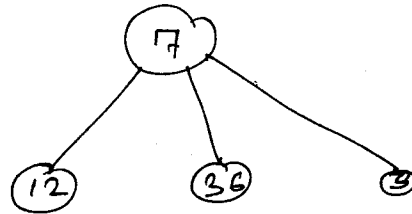
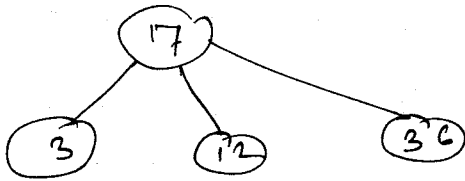
The height of a node is a tree is the number of edges on the longest simple downward path from the node to a leaf.

The height of a tree is the height of its root, which is equal to the largest depth of any node in the tree.

4) Ordered tree:-

A rooted tree in which children of each node are ordered is called an ordered tree. If a node has K children, then there is a 1st child, a 2nd child, a 3rd child $\dots$ a K th child.

eg: The following trees are same when considered as rooted trees but different when considered as ordered trees:-



5) Binary Tree:-

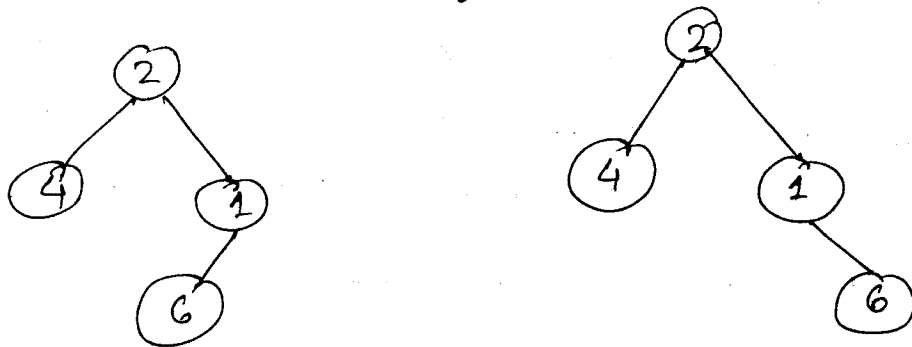
A binary tree either:-

- (i) contains no nodes, or
- (ii) is composed of three disjoint sets of nodes, a root node, a binary tree called its left subtree and a binary tree called its right subtree. 4

The binary tree that contains no nodes is called the empty tree or null tree.

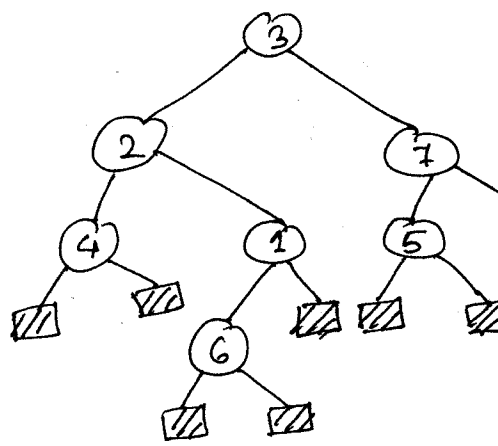
A binary tree is not simply an ordered tree in which each node has degree at max 2. In a binary tree is a node has just one child, the position of the child matters - whether it is the left child or right child.

The following are same when considered as ordered trees but different when considered as binary trees:-



6) Full Binary Tree:- A binary tree in which each node ~~has~~ is either a leaf or has degree exactly 2 is called a full binary tree. There are no degree-1 nodes.

To represent a binary tree as a full binary tree you replace each missing child in the binary tree with a node having no children. These leaf nodes are drawn as squares. eg:-



The original binary tree is represented by the internal nodes of this full binary tree.

7) Positional tree:-

A ~~tree~~ ^{rooted} tree in which the children of a node are labeled with distinct integers is called a positional tree. i.e. the integers provide the position information about the nodes.

A K-ary tree is a positional tree in which every node has a maximum degree of K.

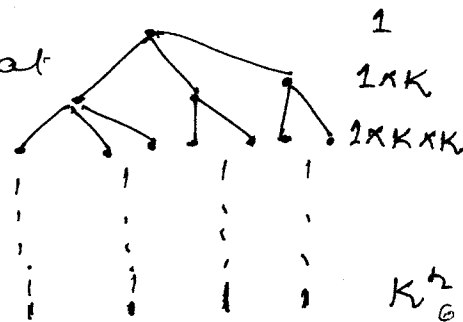
8) Complete K-ary tree:-

A K-ary tree in which all leaves have the same depth and all internal nodes have degree K is called a complete K-ary tree.

How many leaves does a complete K-ary tree of height h have?

The root has K-children at depth 1, each of which has K-children at depth 2 etc.

Thus the no. of leaves = K^h



Height of a complete K-ary tree with n leaves is $\log_K n$. The no. of internal nodes of a complete K-ary tree of height h is:-

$$1 + K + K^2 + \dots + K^{h-1} = \frac{K^h - 1}{K - 1}$$

Thus, a complete binary tree has $2^h - 1$ internal nodes.